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THE LITHIUM PROBLEM WITH IRAM, OSSE, AND *INTEGRAL*

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ABSTRACT

The importance of successful detections of (1) the rotational and vibrational LiH lines at high redshifts, and (2) γ-ray lines from the decays of ^{7}Be, ^{6}Li*, ^{7}Li*, and ^{7}Be* nuclei as diagnostic tools for primordial and galactic nucleosynthesis models is discussed.

Subject headings: nuclear reactions, nucleosynthesis, abundances

1. INTRODUCTION

Lithium is a key element in cosmology, galactic chemical evolution, and stellar models.

According to the standard big bang nucleosynthesis model (SBBN), the predicted primordial abundance of ^{7}Li is within the range 8×10^{-11} to 2×10^{-10} (Walker et al. 1991). The discovery of a uniform surface abundance in the very metal poor (Population II) halo stars (Spite & Spite 1982) at a level of $(^7\mathrm{Li/H})_{\mathrm{Pop\ II}} = (1.2 \pm 0.2) \times 10^{-10}$ has been considered as a confirmation of the predictions of SBBN. Therefore, to explain the observed Population I star ^{7}Li abundance, $(^7\mathrm{Li/H})_{\mathrm{Pop\ I}} \sim 1.2 \times 10^{-9}$, an additional ^{7}Li production is required.

In inhomogeneous big bang nucleosynthesis models (IBBN)—models with inhomogeneous density distribution—the predicted primordial ^{7}Li abundance seems to be at the level of that observed in young Population I stars (Malaney 1991). Explaining the observed Population II stars ^{7}Li abundance would require a uniform and large depletion in old and warm Population II stars.

In addition, no ^{7}Li depletion is predicted in standard (nonrotating) stellar Li depletion models; but recent models with rotation may allow for significant ^{7}Li depletion in Population II stars (Pinsonneault, Deliyannis, & Delmarque 1992; see also Charbonnel, Vauclair, & Zahn 1992). It should be noted that there is no significant ^{6}Li production in SBBN and IBBN models.

For a long time, the galactic cosmic-ray (GCR) interactions with the interstellar medium (ISM) HeCNO nuclei (Population I) have been considered as the major source of ^{6}Li but as a minor source of ^{7}Li (Walker, Mathews, & Viola 1985). Therefore, the lithium abundance origin and evolution may be described by two possible simple scenarios (see Table 1).

In this context, recent observations of lithium abundances have been made for checking some of the above predictions:

1. A measurement of ^{6}Li in the Population II dwarf HD 84937 (Smith, Lambert, & Nissen 1993) is consistent with scenario 1 (Steigman et al. 1993).

2. Measurements of the interstellar ^{7}Li/^{6}Li ratio (Lemoine et al. 1992; Meyer, Hawkins, & Wright 1993) show that the current value of this ratio is equal to or less than its solar system value of 12.5 (Cameron 1982). Such a ratio would require a significant production of ^{6}Li to compete with the ^{7}Li production during the past 4.6 Gyr, since the formation of the solar system.

The purpose of this paper is to explore other Li abundance observations which can be important for our understanding of (1) the primordial nucleosynthesis with a possible discrimination between SBBN and IBBN models, and (2) the early and recent galactic nucleosynthesis with a possible discrimination between GCR origin and "stellar" origin—that is, red giants, novae, supernovae, etc.—for lithium abundance.

Two research programs are suggested with a brief review of their respective contexts: (1) a millimetric search for LiH lines at high redshifts with the IRAM 30 m telescope (§ 2), and (2) a gamma-ray search for recent galactic lithium production with the OSSE telescope borne on the *COMPTON GRO* and with the upcoming *INTEGRAL* mission (§ 3).

2. SEARCH FOR LiH LINES AT HIGH REDSHIFTS

Structure formation scenarios depend on the density fluctuation spectrum at the recombination epoch, $z_r \sim 1100$. Primary temperature fluctuations in the cosmic microwave background (CMB) are a direct probe of these density fluctuations at $z \sim z_r$.

In all structure formation scenarios (see, e.g., Rees 1992) at $z < z_r$, there exists large primordial clouds or protoclouds with mass ranging from $10^6\ M_\odot$ to $10^{15}\ M_\odot$. On the other hand, as the universe expands and cools, at some redshift z, conditions can be met for the formation of molecules such as H_2, HD, LiH etc. Observations of rotational and rotovibrational lines expected to be emitted by these primordial molecules in collapsing protoclouds may constrain the protostructure evolution. Moreover, if these molecules are LiH and LiH^+, observations of their lines may constrain the lithium abundance at high redshifts: a preliminary millimetric search for high-redshift LiH lines is reported in § 2.1. The main features of primordial molecular scattering on CMB photons and suggestions for further observations are given in § 2.2.

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TABLE 1

TWO SCENARIOS FOR LITHIUM ABUNDANCE ORIGIN AND EVOLUTION

	Primordial Lithium	Galactic Lithium Production	Lithium Depletion Model
Scenario 1 (SBBN)	Li^7: 10^{-10}	Li^7: yes	^{7}Li depletion: no
	Li^6: 0	Li^6: yes	In standard model (without rotation)
Scenario 2 (IBBN)	Li^7: 10^{-9}	Li^7: no	^{7}Li depletion: yes
	Li^6: 0	Li^6: yes	In nonstandard model (with rotation)

2.1. *A Very Preliminary Search for LiH Molecules at High Redshifts*

De Bernardis et al. (1993) have searched for molecular lines emitted in collapsing protoclouds by the method proposed by Hogan & Rees (1979) for observations of the redshifted 21 cm line. Several topics are worth mentioning:

1. The rotational and vibrational lines of primordial molecules are observable if (a) The product of molecule abundances times the line strength is high enough; (b) the gas and radiation temperatures are very different; (c) the gas is clumped enough so that differential methods can be used to observe the molecular emission against the background; (d) the optical depth in clouds is high enough.

2. Primordial LiH lines are the most suitable ones for a detection since (a) LiH has a high dipole moment ($\mu_{LiH} = 5.88$ D); (b) LiH can form at $z \leq 200$ where $|T_{gas} - T_{rad}|$ is maximum; (c) LiH abundance—supposed to be formed through radiative association and destroyed by photodissociation—may be greater than 10% of ^{7}Li abundance (see, e.g. Puy et al. 1993 and references therein); (d) LiH rotovibrational lines at 6.5 μm or 1400 cm^{-1}—redshifted for $z \sim 200$ at 1.3 mm or 7 cm^{-1}—fall in the atmospheric window (210–300 GHz or 7–10 cm^{-1}).

3. A high-resolution millimetric telescope is needed since a $D = 2$ kpc cloud, at $z \sim 180$, is observed with an angular size. $\theta = DH_o\Omega z/2c \sim 14\,h$ arcsec if $H_o = 100\,h$ km s^{-1} Mpc^{-1} and the total density of the universe is $\Omega = 1$.

4. The flux in the lines may be estimated through a very simple model: (a) The intensity of the lines depends on three parameters: the cloud density n, the cloud temperature T, LiH abundance X(LiH); (b) In the best case, $\Delta T \sim 1$ mK lines can be expected with a width of $\Delta\nu \sim 0.3$ GHz.

5. A Doppler-induced anisotropy in the 3 K radiation—due to the motion of protoclouds with velocity v—can be estimated by $\Delta T/T \sim \tau v/c$ where $\tau = \tau(n, X(\text{LiH})$ is the optical depth to scattering in the line through the protocloud. In the best case the order of magnitude of the effect is $\Delta T \sim 1$ mK.

6. The IRAM 30 m telescope, with a 12″.5 FWHM beam at 1.3 mm, is well suited for this search. The first millimetric search for high-z LiH lines took place on 1992 April 19 and 20 at Pico Veleta (Spain) for a total observing period of 18 hr. Seven high-latitude regions were observed at $\nu \sim 220$ GHz where two rotovibrational branches of LiH can be observed at $z \sim 180$. No lines were evident in the raw data.

A simple two-parameter linear fit used in the data analysis leads to a 95% upper limit of the order of 20 mK for the intensity of the LiH lines. This leads to (1) An upper limit of the order of 10^7 cm^{-3} for the cloud density if LiH/H $\sim 10^{-12}$ to 10^{-11}, and (2) An upper limit of the order of 500 km s^{-1} for the cloud velocity if the Doppler-induced anisotropy is considered. More observations are needed to improve the sensitivity of this search, that is, to constrain the protocloud evolution and the primordial LiH abundance.

2.2. *Work in Progress and Suggestions for Further Observations*

In order to generalize the Doppler-induced anisotropy case, a detailed study of the role played by primordial molecules on the CMB photons has been undertaken. It seems that:

1. LiH, LiH^+, HeH^+, and HD^+ are the most important primordial molecules. Rotational lines (at $z < 50$) and rotovibrational lines (at $z > 50$) are the main lines. Doppler mechanism is the most efficient and the most probable mechanism for the formation of secondary CMB fluctuations.

2. Primary CMB anisotropies at intermediate angular scales ($\leq 5°$) could be erased by resonant molecular scattering at low frequencies ($v \leq 90$ GHz, $\lambda \geq 3.3$ mm) for which the optical depth τ is much larger than 1. The blurring of primary anisotropies by resonance lines of LiH may occur, at some z, in some frequency bands. The full or the partial blurring of primary fluctuations strongly depends on the abundance of Li. Therefore, the boundary parameters of these bands give an information on the Li abundance and on physical conditions at the redshift z.

3. The same molecules that scatter the CMB photons may induce secondary anisotropies (at small angular scales) if they are condensed into large moving clouds. The amplitude of these anisotropies is expected to be of the order of $\Delta T/T \sim v/c\,\tau \sim 10^{-4}$ for galactic dimensions, decreasing on larger scales.

The following research program is proposed:

1. A search for primary CBR anisotropies at intermediate angular scales in the millimetric region (0.8–2 mm) through the molecular window. Some of us are involved in BOOMERANG, an Antarctica balloon program (Balloon Observation of Millimetric Extragalactic Radiation and Geomagnetics). The first flight is expected in 1995 December. Each flight will have a typical duration of several days, a sky coverage of about 1000 pixels of 20 arcmin^2, and a per pixel sensitivity of 10^{-6}. It would also be important to check whether these CMB anisotropies are erased at radiofrequencies (at $\lambda > 3$ mm where scattering starts to be important). This can be done by a radio receiver having the same angular resolution as BOOMERANG (see Signore, Melchiorri, & Melchiorri 1994).

2. A search for secondary CMB anisotropies at angular scales of galaxies by means of IRAM—10″ of angular resolu-

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tion—taking advantage of the LiH molecule particular spectrum. It is to be noted that observations of LiH clouds are at the sensitivity limit of IRAM since $\Delta T \sim 0.1$–0.3 mK is to be reached.

These ground-based and balloon-borne experiments (devoted to observations of anisotropies) will set, in the near future, a detection of, or at least, new constraints on the ^{7}Li abundance at a high redshift and will allow for a reliable discrimination between SBBN or IBBN models.

3. SEARCH FOR GALACTIC γ-RAY LINES FROM ^{7}Be, ^{7}Be*, ^{6}Li*, and ^{7}Li* NUCLEI

The *Compton Gamma Ray Observatory* (*CGRO*) and, in the near future, *INTEGRAL* (*International Gamma Ray Astrophysics Laboratory*) make it possible to check the existence of some recent "stellar" lithium sources and also the following:

1. The ν-process in Type-II supernova models (Woosley et al. 1990) and the nova theory (see Harris, Leising, & Share 1991 and references therein) in observing the 0.478 MeV γ-ray line emitted as ^{7}Be decays to ^{7}Li in their expanding ejecta.

2. α-α collision scenarios on collapsed stars—such as accreting neutron stars, newly born pulsars—in observing the 0.429, 0.478, 3.56 MeV γ-ray lines emitted from ^{6}Li*, ^{7}Li*, ^{7}Be* nuclei produced in their surroundings (e.g., Burdyzha, Lipunov, & Yudin 1991).

For *case* 1, the anticipated fluxes are such that:

OSSE—with a sensitivity of $3 \times 10^{-5}\ \gamma\ \mathrm{cm}^{-2}\ \mathrm{s}^{-1}$ for a line at 0.5 MeV—can detect the 0.478 MeV line if a nova is at 1.5 kpc, a (15–25) $M_\odot$ SN II is at 20 kpc, and a 35 $M_\odot$ SN II is at 26 kpc.

INTEGRAL—with a sensitivity of $3 \times 10^{-6}\ \gamma\ \mathrm{cm}^{-2}\ \mathrm{s}^{-1}$—will detect the 0.478 MeV line if a nova is at 5 kpc, a (15–25) $M_\odot$ SN II is at 60 kpc, and a 35 $M_\odot$ SN II is at 82 kpc! Let us only recall that the great SN 1987A is at 55 kpc!

A remark about Type-II supernovae should be made. Because ^{7}Be and ^{7}Li are not produced by spallation but by the neutrino-process, the same ^{7}Li production is expected in stars of zero metallicity, solar metallicity Z_O, 2 Z_O (S. Woosley 1992, private communication). Accordingly: (1) the ^{7}Li abundance observed in the solar neighborhood (Cameron 1982) can be accounted for and (2) the galactic ^{7}Li enrichment would be predicted to occur rather early in the galactic history if indeed a substantial fraction of its ^{7}Li originates from SNs II through the ν-process (Signore & Vedrenne 1993). Therefore, SNs II should play a "target of opportunity" role for *CGRO* and *INTEGRAL*: the ^{7}Li flux illuminates not only the ν-process but also the history of the Galaxy!

For the second case of α-α collisions on collapsed objects located at some 5 kpc, OSSE would be able to detect some or all the 0.429, 0.478, 3.56 MeV lines for a high concentration of α-particles in the accreted matter onto, or in the ejected matter from, the collapsed object. *INTEGRAL* will be able to detect all these lines for a "normal" abundance of α-particles in the vicinity of the collapsed object.

A further remark on these scenarios: The requirement relating to the abundance in α-particles is difficult to justify. Accordingly, contributions of these processes to the galactic productions of ^{6}Li and ^{7}Li remain difficult to be quantitatively evaluated. Nuclear γ-ray line observations would then be clear-cut tests for such scenarios.

In any case, a successful detection of γ-ray lines of ^{7}Be nuclei from novae and SNs II and/or ^{6}Li*, ^{7}Li*, ^{7}Be* nuclei from neutron stars would yield invaluable information on the galactic sources of ^{7}Li and ^{6}Li: a major input of chemical evolution models of the Galaxy, generally considered as an adjustable parameter, would be physically determined!

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